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SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, INTERLINING, COTTON WARP, AND SPUN HAIR-WOOL OR RAYON FILLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for interlining cloth constructed from cotton warp yarns and a blend of spun goat-hair and wool or spun goat-hair and rayon filling yarn.

1.2 Classification. The cloth shall be of the following types and classes as specified (see 6.2):

Type I	-	Medium weight
Type II	-	Heavy weight
Type III	-	Light weight
Class 1	-	Natural
Class 2	-	Dyed

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATION

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Cotton (warp). The cotton shall be carded, drawn and spun into either a 2 ply or singles yarn.

3.2.2 Goat-hair and wool (filling). The goat-hair and wool shall be blended together and then spun on the worsted system into a singles yarn. The wool shall be of staple fleece or pulled wool. The use of reprocessed or reused wool is prohibited.

* 3.2.2.1 Animal fiber content (hair-wool filling). The cloth shall analyze no less than 50 percent animal fiber based on the dry desized weight when tested as specified in 4.4.

* 3.2.3 Goat-hair and rayon (filling). The goat-hair and rayon shall be blended together and spun into singles yarn.

* 3.2.3.1 Animal fiber content (hair-rayon filling). The cloth shall analyze no less than 32 percent animal fiber based on the dry desized weight when tested as specified in 4.4.

3.3 Color. The color of the class 2 cloth shall be Black 341 (see 6.4), and shall match the standard sample (see 6.3).

* 3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

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* 3.3.2 Matching. The color of the dyed cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.3 Colorfastness. The dyed and finished cloth shall show fastness to wet drycleaning, perspiration and steam pressing equal to or better than the standard sample when tested as specified in 4.4. When no standard sample has been established or designated as applicable to colorfastness, the dyed cloth shall show "good" colorfastness to wet drycleaning, perspiration and steam pressing when tested as specified in 4.4.

3.4 Physical requirements. The physical requirements of the finished cloth shall be as specified in table I when tested as specified in 4.4.

* TABLE I. Physical requirements

Type	Weight, ounces per sq. yd.		Yarns per inch (minimum)		Breaking strength (minimum) pounds		Stiffness 1/ (load-pounds)
	Min.	Max.	Warp	Filling	Warp	Filling	Min.
I	7.0	8.0	42	36	50	45	0.011
II	7.5	9.0	42	36	50	45	0.016
III	5.0	6.5	42	36	50	45	0.005

1/ Measured parallel with filling direction.

3.4.1 Weave. The weave shall be plain.

3.4.2 Width. The width shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish. The class 1 cloth shall have an unbleached natural finish. The class 2 cloth shall be dyed.

3.5.1 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 8.0 percent when tested as specified in 4.4.

* 3.5.2 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.

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3.6 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 3.0 percent in the warp direction nor more than 1.0 percent in the filling direction when tested as specified in 4.4. The preshrinkage processes used shall not be identified by name or trademark on the cloth, ticket or package.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put-up in full width rolls as specified in PPP-P-1134.

3.8 Wool content label. The wool cloth manufactured under this specification shall be labeled on the ticket in accordance with the Wool Products Labeling Act of 1939.

3.9 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.5.

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* 4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 yard, full width of the finished cloth. The acceptable quality level shall be 6.5 defects per 100 units (yards). The sample size shall be in accordance with inspection level I of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with table II. An approximate equal number of yards shall be examined from each roll selected.

Defects

Hole, cut or tear
Smash, multiple floats, spots or stains
Embedded crease or wrinkle
Broken or missing pick or end, forming heavily jerked-in area or missing for 18 inches or more, any multiple

* 4.2.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects

Width less than specified 1/
Overall uncleanness 1/
Baggy, ridgy or wavy cloth

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

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TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	1
35,001 up to and including 150,000	20	2
150,001 and over	32	3

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.2.2.4 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.5 Examination for identification of preshrinkage process and for noncompliance with the Wool Products Labeling Act. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Wool Products Labeling Act.

4.3 Packaging inspection. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with section 5 requirements.

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* 4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width, of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Warp yarn:		
Cotton identification	3.2.1	1200 1/
Filling yarn:		
Spun goat-hair and wool identification	3.2.2	1/
Spun goat-hair and rayon identification	3.2.3	1/
The wool shall contain no reprocessed or reused wool	3.2.2	1/
Yarn (singles)	3.2.2 and 3.2.3	1/
Animal fiber content	3.2.2.1 and 3.2.3.1	2100
Presence of labile sulfur	3.3.1	2020
Colorfastness to:		
Wet drycleaning	3.3.3	5622
Perspiration	3.3.3	5680
Steam pressing	3.3.3	5640
Weight	3.4	5041

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TABLE III. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Stiffness (filling only)	3.4	5202
Weave	3.4.1	Visual 2/
Nonfibrous materials	3.5.1	2611
pH	3.5.2	2811
Shrinkage	3.6	5558 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the results shall be reported as "pass" or "fail".

3/ Method 5558 shall be used except that the specimens shall be dried as specified in Method 5550. The cloth shall be kept submerged for a period of 30 minutes in water containing a minimum amount of a suitable wetting agent to thoroughly wet out the fabric.

* 5. PACKAGING

5.1 Preservation-packaging. Put-up shall be level A in accordance with the applicable requirements of PPP-P-1134.

* 5.2 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

* 5.2.1 Level A. The cloth shall be packaged in accordance with the applicable requirements of PPP-P-1134.

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* 5.2.2 Commercial packaging. The cloth shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.3 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.3.1 Levels A and B. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3.2 Commercial packing. The cloth preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.4 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with the requirements of PPP-P-1134. Marking for commercial shipments shall be the same as that specified for level C.

6. NOTES

6.1 Intended use. The cloth is intended for use as an interlining in clothing items.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Width of cloth required (see 3.4.2).
(d) Length required when other than specified (see 3.7).
(e) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

* 6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

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* 6.4 Dyestuff formulation, Black 341. A suggested but not mandatory formulation for Black 341 is as follows:

Direct Black RL
Neutral Premetallized Black C-BL
Aftertreated with copper sulfate

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD
Navy - MC

Preparing activity:

Army - GL

Project No. 8305-0556

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